22CH203 – ENGINEERING CHEMISTRY II

UNIT V: NUCLEAR REACTIONS

Lesson Plan 3: Nuclear reactions: Fission and fusion - Radiocarbon dating

Topics: Applications of nuclear fission, Nuclear reactor or atomic pile, Radiocarbon dating

1. APPLICATIONS OF NUCLEAR FISSION:

A heavy isotope like uranium (^{235}U) or plutonium (^{239}Pu) can undergo nuclear chain reaction with the liberation of large amount of energy. The energy released by the fission of nuclei is called nuclear energy. The fission of ^{235}U or ^{239}Pu occurs instantaneously producing vast quantities of energy.

If the reaction is uncontrolled, it is accompanied by explosive violence and can be used in atom bombs. When the reaction is controlled in a reactor, the fission reaction is harnessed to produce electricity.

1. **Electricity Generation:** Nuclear fission powers plants to create electricity without releasing a lot of pollution.
2. **Nuclear Weapons:** Fission reactions create powerful explosions, which is how nuclear bombs work.
3. **Radioisotope Production:** Fission helps make special materials used in medicine, industry, and research.
4. **Nuclear Research:** Scientists use fission to learn more about how atoms work and test new ideas.
5. **Propulsion Systems:** Fission can power engines in submarines, spaceships, and maybe even future cars and trains, giving them lots of energy for long trips.
6. **Desalination:** Fission reactors can help turn seawater into fresh water, which is useful in places where there isn't much clean water available.

2. NUCLEAR REACTOR or ATOMIC PILE

A nuclear reactor is a device in which the nuclear fission reaction is carried out at a controlled rate so that the liberated energy can be utilized for constructive purposes like generation of electricity.

Components of a Nuclear Power Reactor

The chief components of a nuclear reactor are as follows:

The reactor core is the principal component of a nuclear reactor. It is that part where controlled fission reaction is made to occur and where heat energy is liberated. It consists of an assembly of fuel elements, moderator, coolant and control rods.

2.1. *Fuel element:* The commonly used fissionable materials are as follows:

(a) Natural uranium which contains 0.71% of fissile 235U isotope

(b) Enriched uranium in which 235U has been increased to a few per cent

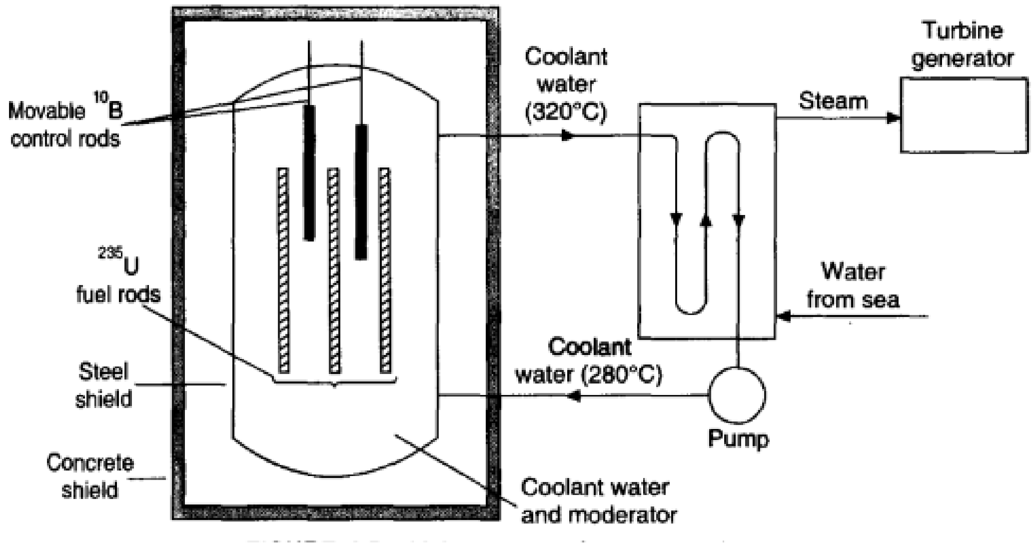
2.2. *Moderators:* The function of the moderator is to slow down the highly energetic neutrons produced in the fission reaction. Heavy water (D20), graphite, beryllium, etc. are used as moderators. Neutrons slow down by losing the energy due to collision with atoms or molecules of the moderator.

2.3. *Coolants:* The function of a coolant is to remove the intense heat produced in the reactor. Through the heat exchangers, the coolants transfer the heat to the secondary thermal system of the reactor. Ordinary water, heavy water, liquid metal (sodium), etc. are used as coolants.

2.4. *Control rods:* The chain reactions are controlled by using movable rods made up of boron or cadmium. These rods are having high absorption capacity for neutrons. If the control rods are pushed into the reactor core, they absorb more neutrons and cut down the reactivity. If the rods are raised, they will absorb less neutrons and the reaction will be very fast.

3. Light Water Nuclear Power Plant

Most commercial nuclear power plants today are light water reactors. As the fuel rods are submerged in water, this is called a light water power plant (Figure 4.5). Here the water acts as a coolant as well as a moderator.



Light water nuclear power plant

The fission reaction is controlled by inserting or removing the control rods automatically from spaces in between the fuel rods. The heat emitted by fission of ^{235}U in the fuel core is absorbed by the coolant (light water). The heated coolant (water at 320°C) then goes to the heat exchanger. Here the coolant transfers the heat to sea water which is converted into steam. The steam then drives the turbines generating electricity.

The nuclear waste obtained from the reactor is packed in concrete barrels which are buried deep in the earth or dumped in the sea.

4. RADIOCARBON DATING:

Radioactive dating is a process by which the approximate age of an object is determined through the use of certain radioactive nuclides. For example, carbon-14 has a half-life of 5,730 years and is used to measure the age of organic material. The ratio of carbon-14 to carbon-12 in living things remains constant while the organism is alive because fresh carbon-14 is entering the organism whenever it consumes nutrients. When the organism dies, this consumption stops, and no new carbon-14 is added to the organism. As time goes by, the ratio of carbon-14 to carbon-12 in the organism gradually declines, because carbon-14 radioactively decays while carbon-12 is stable. Analysis of this ratio allows archaeologists to estimate the age of organisms that were alive many thousands of years ago.

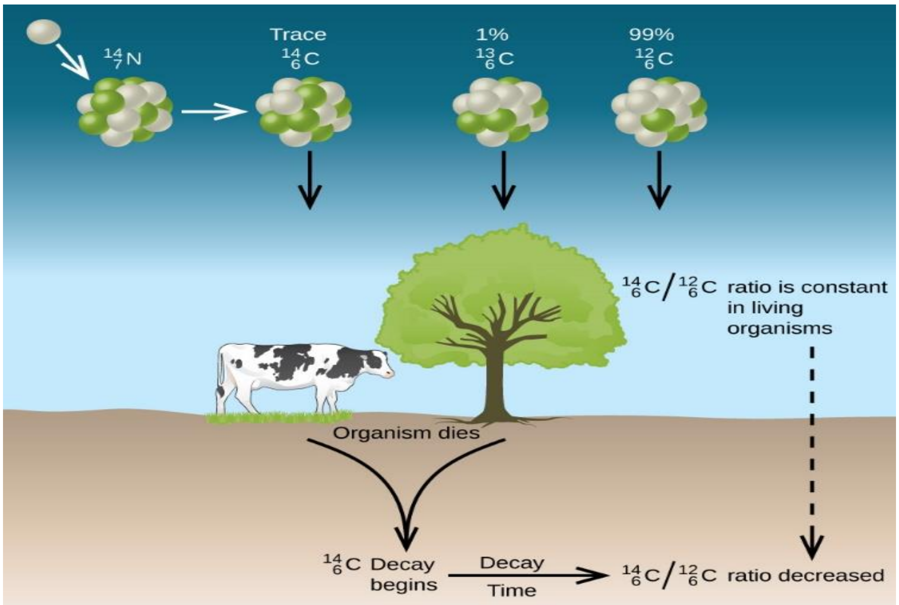


Figure. Along with stable carbon-12, radioactive carbon-14 is taken in by plants and animals, and remains at a constant level within them while they are alive. After death, the C-14 decays and the C-14:C-12 ratio in the remains decreases. Comparing this ratio to the C-14:C-12 ratio in living organisms allows us to determine how long ago the organism lived (and died).

C-14 dating does have limitations. For example, a sample can be C-14 dating if it is approximately 100 to 50,000 years old. Before or after this range, there is too little of the isotope to be detected. Substances must have obtained C-14 from the atmosphere. For this reason, aquatic samples cannot be effectively C-14 dated. Lastly, accuracy of C-14 dating has been affected by atmosphere nuclear weapons testing. Fission bombs ignite to produce more C-14 artificially. Samples tested during and after this period must be checked against another method of dating (isotopic or tree rings).

To calculate the age of a substance using isotopic dating, use the equation below:

how old (time)=$n \times t_{1/2}$

where n is the number of half-lives and $t_{1/2}$ is the half-life (in time).

SAMPLE PROBLEM

How long will it take for 18.0 grams of Ra-226 to decay to leave a total of 2.25 grams? Ra-226 has a half-life of 1600 years.

Solution

18.0g ⇒ 9.0g ⇒ 4.5g ⇒ 2.25g, this is three half-lives

$$\text{how old (time)} = 3 \times 1600\, years$$

This decay process takes 4800 years to occur.

Discussion Questions:

- Predict the effects of selecting fuel, such as uranium or plutonium, on reactor efficiency and the generation of waste in nuclear fission reactors.
- How do neutrons contribute to nuclear reactions, and what methods are employed to regulate or influence their behavior in different nuclear processes?
- How do control rods function in regulating the rate of fission reactions within a nuclear reactor, and what happens if they fail?
- The nuclear power plants in Tamil Nadu, situated at Kalpakkam and Kudankulam, are positioned close to the sea. Provide your rationale.